FREDDY ROJAS

LinkedIn | freddy.indevs.in | +51956957062 | Efreddyrl@gmail.com | GitHub

Skills

- **Programming languages**: Golang | Python | Javascript, TypeScript, Node.js
- **Backend technologies**: Gorilla/mux, Gin, | Flask, FastAPI, Django, Express, Hapi, Koa.js, NestJS | RESTful APIs, GraphQL, gRPC, WebSockets.
- **AI, LLMs & Low-Code/No-Code**: OpenAI, Gemini, Ollama | RAG | CAG | LightRAG | AnythingLLM | LiteLLM | LangChain | LangGraph | LangSmith | n8n | Streamlit | ChromaDB | PGvector
- **DevOps tools**: AWS/AZURE/GCP, Docker, Kubernetes, CI/CD (Jenkins/GitHub Actions), Terraform
- **Database management systems**: PostgreSQL, MySQL, DynamoDB, Redis (cache and sessions), MongoDB, SQLC
- **Frontend technologies**: Angular | React, Next.js | Vue | CSS, Tailwind CSS, Bootstrap
- **Architecture and Design**: Object-Oriented Programming (OOP) | Functional Programming | Design Patterns, Data Structures, Algorithms | Domain-Driven Design (DDD) | Test-Driven Development (TDD) | Hexagonal Architecture | Multithreaded and Asynchronous Programming
- **Automation & Testing**: Selenium, Puppeteer, Playwright, Requests, Cheerio, HTTP, BeautifulSoup, Fetch, Axios.

Experience

Software and AI Engineer | **GLOBAL STREAMING SOLUTIONS** | **Remote** | **01/2024 - Now**

- **HIGH AVAILABILITY POS SYSTEM (GOLANG/REACT)**
  - Hexagonal Backend (Golang): Implementation with Gorilla/Mux and PostgreSQL (SQLC), achieving a 50-70% reduction in RAM consumption vs. traditional ORMs.
  - Security & RBAC: Granular access control design with Casbin, managing dynamic roles and adaptive policies.
  - Modular Frontend: SPA with React + TypeScript, using Zustand for state management and React Query for caching.
  - DevOps: CI/CD with Jenkins and deployment on AWS/AZURE/GCP/DigitalOcean via Nginx reverse proxy.
- **AI ARCHITECTURE & AUTONOMOUS AGENTS**
  - RAG & API Pipelines: Scalable design on private data using on-premises LLMs (Ollama) and inference exposure using high-performance microservices in FastAPI.
  - Autonomous Agents (Python): Agent development (LangChain, LangGraph) with "Tool Use" to orchestrate APIs and process documentation in real time.

Software Engineer | **BLUECODE** | **Remote** | **11/2021 - 12/2023**

- Architected and led the data acquisition engine for Prime.Scout, a Delaware-based Price Intelligence startup.
- Developed scraping bots for ecommerce using TypeScript/Node.js (with Axios, Fetch, Request) and Python/Golang (with native HTTP library), optimizing the extraction of product, price and catalog data.
- Migrated and improved legacy infrastructure by replacing Puppeteer/Playwright with solutions based on HTTP Direct (Node.js+Go), reducing resource consumption by 90% and accelerating response time by 95%.
- Implemented a scalable URL management system: from initial website crawling to structured data extraction (JSON/CSV), storing results in PostgreSQL databases with schema normalization.
- Containerized bots with Docker to ensure portability, execution in cloud environments (AWS EC2) and Automatic scaling on demand.
- Created a notification bot in Telegram (TypeScript/Node.js) for real-time monitoring, crash alerts and daily reports of extracted data.
- Development of Internal Tools (React, Vue).

Freelance Software Developer | **GLOBAL STREAMING SOLUTIONS** | **Remote** | **01/2018 - 11/2021**

- B2B SaaS Development & Product Lead (React, Vue.js & Node.js): Architected and launched a proprietary SaaS platform, successfully scaling it to +20 active paying B2B clients. Built responsive SPAs using React and Vue.js for complex real-time interactions, and provided maintenance for legacy Angular modules.
- Web Scraping & Data Mining (Python/Django): Developed robust data extraction pipelines using Selenium, Scrapy, BeautifulSoup, and Requests to bypass anti-bot mechanisms. Built custom data management dashboards in Django to process and structure massive datasets at scale.
- Frontend Modernization (Next.js): Migrated legacy static views to Next.js, significantly modernizing the frontend stack, improving SEO performance, and streamlining content management workflows.

Education

Software Engineering | **Technological University of Peru** | **Remote** | **Ongoing**

Portfolio

- **Portfolio: freddy.indevs.in**
- **API Hexagonal & CQRS | FastAPI, Docker, PostgreSQL**
  Advanced microservices architecture with separation of layers and design patterns. Focused on scalability and maintainability.
  github.com/RLFreddy/hexacqrs-starter
- **Epayco Virtual Wallet Simulation | NestJS, Vue.js, Microservices**
  A complete full-stack system with containerized microservices architecture and real-time transaction flow.
  github.com/RLFreddy/Epayco-wallet
- **High Availability POS System | Golang, React, SQLC**
  Development of an optimized point of sale (POS) under hexagonal architecture to ensure low resource consumption and high concurrency. (Demo in local)
- **Advanced Scraping Motors | Python, Go, Node.js, Playwright**
  - https://drive.google.com/file/d/1u8vHHuXCseTaFKW5Z-BzWteJJ_KoMljG/view?usp=sharing
  - https://drive.google.com/file/d/1W2GWeLivyIG8g9yA4S9sMcf8JZ7F6-TW/view?usp=sharing